

Module 10 – Capstone Project

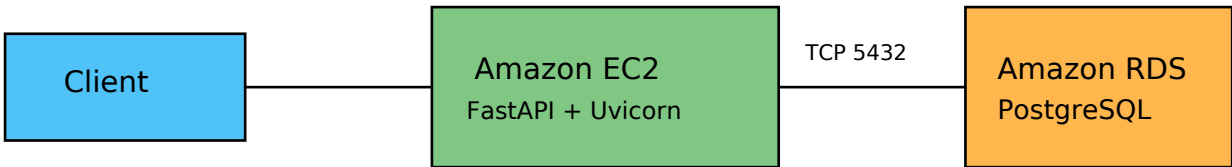
Part 4 – Cloud Infrastructure Deployment Report

Trello REST API Backend

Project Overview

The Trello Backend REST API was deployed on Amazon Web Services (AWS). The backend application is hosted on Amazon EC2 and securely connected to an Amazon RDS PostgreSQL database. Database schema creation and updates are managed using Alembic migrations. Deployment was validated by successfully connecting to the database and performing API operations.

System Architecture



AWS Resources Used

AWS Service	Purpose
Amazon EC2	Hosts FastAPI backend
Amazon RDS PostgreSQL	Managed relational database
Amazon VPC	Private networking
Security Groups	Secure communication between EC2 and RDS
Alembic	Database schema migrations

Deployment Summary

- Amazon EC2 instance provisioned for backend deployment.
- Amazon RDS PostgreSQL database configured.
- Security Groups configured for secure database access.
- Backend environment updated with RDS endpoint.
- Database connectivity verified using psql.
- All Alembic migrations executed successfully.
- API tested successfully with user registration stored in Amazon RDS.

Deployment Status

Component	Status
EC2 Backend	Completed
Amazon RDS	Completed
Database Connectivity	Verified
Alembic Migrations	Completed
API Validation	Successful

Conclusion

The cloud deployment demonstrates a working AWS-hosted backend architecture with Amazon EC2, Amazon RDS PostgreSQL, secure networking through Security Groups, and successful database migration using Alembic. The application was validated through successful API execution and database persistence.